

**PARUL UNIVERSITY**  
**FACULTY OF ENGINEERING & TECHNOLOGY**  
**B.Tech., Summer 2017 - 18 Examination**

Semester: 5

Subject Code: 03105302

Subject Name: Design and Analysis of Algorithm

Date: 31/05/2018

Time: 10:30 am to 1:00 pm

Total Marks: 60

**Instructions:**

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

**Q.1 Objective Type Questions - (All are compulsory) (Each of one mark)****(15)**

1. \_\_\_\_\_ is the first step in solving the problem.
 

|                              |                         |
|------------------------------|-------------------------|
| A. Understanding the Problem | B. Identify the Problem |
| C. Evaluate the Solution     | D. None of these        |
2. Examples of  $O(n^2)$  algorithms are \_\_\_\_\_.
 

|                           |                                                |
|---------------------------|------------------------------------------------|
| A. Adding of two Matrices | B. Initializing all elements of matrix by zero |
| C. Both A and B           | D. Neither A nor B                             |
3. Express the formula  $(n-1)*(n-5)$  in terms of big Oh notation.
 

|           |                |           |             |
|-----------|----------------|-----------|-------------|
| A. $O(1)$ | B. $O(\log n)$ | C. $O(n)$ | D. $O(n^2)$ |
|-----------|----------------|-----------|-------------|
4. The algorithm like Quick sort does not require extra memory for carrying out the sorting procedure. This technique is called \_\_\_\_\_.
 

|             |           |             |                 |
|-------------|-----------|-------------|-----------------|
| A. in-place | B. stable | C. unstable | D. in-partition |
|-------------|-----------|-------------|-----------------|
5. The recurrence relation capturing the optimal execution time of the Towers of Hanoi problem with  $n$  discs is ?
 

|                         |                         |
|-------------------------|-------------------------|
| A. $T(n) = 2T(n-2) + 2$ | B. $T(n) = 2T(n-1) + n$ |
| C. $T(n) = 2T(n/2) + 1$ | D. $T(n) = 2T(n-1) + 1$ |
6. A \_\_\_\_\_ sort compares adjacent elements in a list and switches where necessary.
7. The number of swapping needed to sort the numbers 8, 22, 7, 9, 31, 19 in ascending order, using bubble sort is \_\_\_\_\_.
8. In a Max heap the largest key is at \_\_\_\_\_. (Root/Leaf/Node)
9. Evaluate  $5*(x+y)-4*y/(z+6)$  where  $x = 2$ ,  $y = 3$ , and  $z = 6$ .
10. Evaluate  $a-2>b$  where  $a=6$ ,  $b = 8$ .
11.  $f_1(n) = 2^n$ ,  $f_2(n) = n^{3/2}$ ,  $f_3(n) = n \log n$ ,  $f_4(n) = n^{\log n}$  arrange these functions in ascending order of growth rate.
12. What will be the complexity of the following code snippet?

```
void function(int n)
{
    for(int i=0;i<9999;i++)
    {
        printf("focus");
    }
}
```

```

13. void function(int n)
{
    for(int i=1;i<n;i=i*2)
    {
        printf("Design");
    }
}

```

```

14. void function(int n)
{
    for(int i=1;i*i<n;i++)
    {
        printf("organization");
    }
}

```

```

15. void function(int n)
{
    int n = 2^(2^k);
    for(int i=1;i<=n;i++)
    {
        j=2;
        while(j<=n)
        {
            j = j*j;
            printf("algorithms");
        }
    }
}

```

**Q.2 Answer the following questions. (Attempt any three) (15)**

- A) What is Principle of Optimality in Dynamic programming?
- B) What is Recursion? Give one example.
- C) What is algorithm? Write down the characteristics of an algorithm.
- D) Name all asymptotic notations used in algorithm analysis with their significance.

**Q.3** A) We have a knapsack with max weight capacity  $W = 5$ . Fill the knapsack with items such that the benefit (value or profit) is maximum using dynamic programming technique with the following table data which contains the items along with their value and weight. **(07)**

| Item (i)  | 1   | 2  | 3  | 4  |
|-----------|-----|----|----|----|
| value (v) | 100 | 20 | 60 | 40 |
| weight(w) | 3   | 2  | 4  | 1  |

- B) Find Minimum Spanning Tree for the given graph with Prim's algorithm (initialization from node A) using given Figure1. (08)

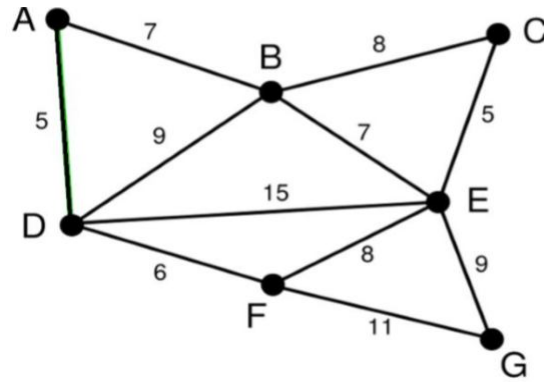


Figure 1 Find MST

OR

- B) Find Minimum Spanning Tree for the given graph with Kruskal's algorithm (initialization from node A) using above given Figure1. (08)

- Q.4 A) Explain divide and conquer concept with any suitable algorithm as an example. (07)

OR

- A) Explain making change coin problem with any suitable example. (07)  
 B) A travelling salesman problem is given in the following figure 2. Find out minimum cost path. (08)

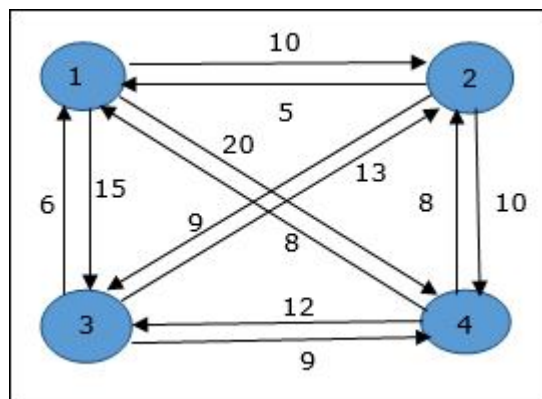


Figure 2 Travelling Salesman Problem